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English Due Oct 25, 11:59 PM IST English

5.2 2x2 Image Transformations Self-check Quiz

coach

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Assignment details

Due Oct 25, 11:59 PM IST   **Attempts** Unlimited

Submitted Oct 20, 2:54 AM IST

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1. What types of transformations **cannot** be represented with a 2x2 matrix?   **1 / 1 point**

- (●) Translation
- ( ) Scaling
- ( ) Skewing
- ( ) Rotation

Correct
A translation *cannot* be represented by a 2x2 matrix. To see why this is so we must realize that a 2x2 matrix is actually generating a *linear combination* of its inputs

$$[x', y'] = [x, y]\begin{bmatrix} a & b \\ c & d \end{bmatrix} = [ax + by, cx + dy]$$

In such a combination, there is no way to represent a term which is independent of both x and y.

2. Image Filtering and Image Warping act on the domain and the range of the image respectively.   **1 / 1 point**

Determine if the above statement is True/False?

- ( ) True
- (●) False

Correct
Filtering acts on the brightness of the pixels in the image (the range of the image) while Warping changes the shape of the image (the domain).